

Certain low confidence words experiment by GPT-3.5-Turbo

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Idea

In this step, Enno suggested me to ask ChatGPT to only fix certain low-confidence words in the ASR output rather than the whole sentence.

For this experiment, I use GPT-3.5-Turbo 1106 as a language model to correct the ASR errors.

Here is the prompt of this experiment

```
{  'role': 'system',  
  'content': f"""You are a helpful assistant that corrects ASR errors. \  
  
You will be presented with an ASR transcription and low confidence words in that transcription. \  
  
the input will be formatted as json with keys text and low_confidence_words,\  
  
where the text is the ASR transcription and low_confidence_words contains the list of words with low asr confidence. \  
  
your task is to check if the low confidence words make sense in the transcription and if they do not, replace them with better words. \  
  
provide your output as a string. \  
  
Do not change the case, for example, lower case or upper case, in the transcription. \  
  
Do not output any additional text that is not the corrected transcription. \  
  
Do not write any explanatory text that is not the corrected transcription.  
"""}

```

Providing example in prompt

```
{  
    'role': 'user',  
    'content': '{"text": "Why not allow your silver tuff to luxuriate in a natural manner?",  
"low_confidence_words":["tuff"]}'  
},  
{'role': 'assistant', 'content': "why not allow your silver tufts to luxuriate in a natural manner?"},  
{'role': 'user', 'content': json.dumps(asr_transcription)}  
}
```

Code snippet

```
for i,d in enumerate(data):
```

```
    asr_transcription = confidence_score_word_level(d["asr_transcription"], confidence = True)
```

```
    low_confidence_words = [word["text"] for word in asr_transcription["words"]
```

```
        if word["confidence"] <= THRESH]
```

```
    if len(low_confidence_words) > 0:
```

```
        asr_transcription = {"text": asr_transcription["text"], "low_confidence_words": low_confidence_words}
```

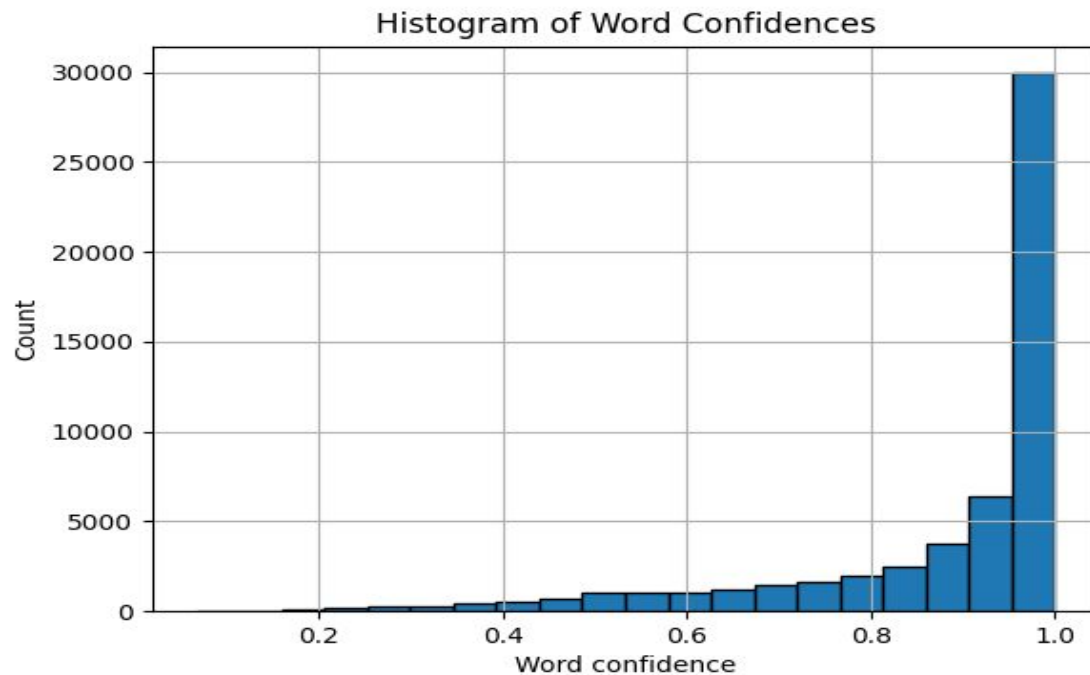
```
        reference_transcription= d["reference_transcription"]
```

```
        data1.append({"asr_transcription": asr_transcription, "reference_transcription": reference_transcription})
```

Here is the one example of output for this experiment

```
{  
  "asr_transcription": {  
    "text": " But, Dukalion and Piro were very sad. For they knew that they were the only persons who were left alive in all the land.",  
    "low_confidence_words": [  
      "Dukalion",  
      "Piro"  
    ]  
  },  
  "reference_transcription": "BUT DEUCALION AND PYRRHA WERE VERY SAD FOR THEY KNEW THAT THEY WERE THE ONLY PERSONS WHO WERE LEFT ALIVE IN ALL THE LAND",  
  "corrected_asr_transcription": "But, Deucalion and Pyrrha were very sad. For they knew that they were the only persons who were left alive in all the land."  
},
```

Histogram of word confidence



Results for this experiment by GPT-3.5 Turbo (Thresh =0.5)

```
WER_original:      8.6116%  
WER_corrected_chatgpt: 7.5211%  
CER_original:      3.5707%  
CER_corrected_chatgpt: 3.6174%  
SER_original:      61.4502%  
SER_corrected_chatgpt: 59.1565%  
---
```


Next step

Now based on histogram, we should define the range of Thresh and call the chatgpt for this range.

And finally compare the errors of each thresh together.